

Performance Testing Report

Project Details

Item	Details
Project Name	AI/ML Web Application
Testing Type	Performance Testing
Tool Used	Apache JMeter
Test Date	18 June 2026
Tester	QA Tester Intern
Environment	Windows 11, Chrome Browser

Objective

To evaluate the application's responsiveness, stability, scalability, and reliability under different user loads.

Test Scenarios

Scenario ID	Scenario	Concurrent Users
PT-001	User Login Load Test	100
PT-002	Dashboard Access Load Test	200
PT-003	File Upload Performance Test	100
PT-004	Prediction Generation Load Test	50
PT-005	Report Download Test	150
PT-006	API Stress Test	500 Requests

Test Execution Results

PT-001: Login Load Test

Metric	Result
Virtual Users	100
Average Response Time	1.1 sec
Throughput	180 req/sec
Error Rate	0%
Status	Pass

Observation: System handled 100 concurrent login requests successfully

PT-002: Dashboard Load Test

Metric	Result
Virtual Users	200
Average Response Time	1.8 sec
Throughput	240 req/sec
Error Rate	0%
Status	Pass

Observation

Dashboard remained responsive under expected load.

PT-003: File Upload Performance Test

Metric	Result
Virtual Users	100
Average Upload Time	2.5 sec
Maximum Time	4.2 sec
Error Rate	1%
Status	Pass

Observation

Large file uploads completed successfully with minimal failures.

PT-004: Prediction Generation Test

Metric	Result
Virtual Users	50
Average Response Time	3.4 sec
Maximum Response Time	5.1 sec
Error Rate	0%
Status	Pass

Observation

AI/ML prediction engine performed within acceptable limits.

PT-005: Report Download Test

Metric	Result
Virtual Users	150
Average Download Time	1.6 sec
Throughput	220 req/sec
Error Rate	0%
Status	Pass

Observation

Reports downloaded successfully under concurrent access.

PT-006: API Stress Test

Metric	Result
Requests Sent	500
Successful Requests	492
Failed Requests	8
Average Response Time	1.9 sec
Error Rate	1.6%
Status	Pass with Observation

Observation

Minor failures observed during peak load but system recovered automatically.

Resource Utilization

Resource	Average Usage
CPU Usage	62%
Memory Usage	68%
Disk Usage	45%
Network Usage	54%

Result

System resources remained within acceptable limits.

Performance Test Summary

Metric	Value
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Total Scenarios Executed	6
Passed	5
Passed with Observation	1
Failed	0
Peak Users Tested	200
Maximum Throughput	240 req/sec
Average Response Time	2.05 sec
Highest Error Rate	1.6%

Recommendations

1. Implement load balancing for high-traffic scenarios.
 2. Optimize database queries to reduce response times.
 3. Introduce API caching for frequently accessed endpoints.
 4. Monitor resource utilization during production deployment.
 5. Conduct stress testing with 500+ concurrent users in future cycle
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Final Conclusion

- ✓ Application demonstrated stable performance under expected workload conditions.
- ✓ Average response time remained below the target threshold of 3 seconds for most operations.
- ✓ No critical performance bottlenecks were identified.
- ✓ The application is Performance Test Approved and recommended for release.

QA Sign-Off

Item	Status
Load Testing	Passed
Stress Testing	Passed
Resource Monitoring	Passed
Performance Validation	Approved

Prepared By: QA Tester Intern

Document Version: 1.0

Status: Performance Testing Completed Successfully